

Inference-Time Decoding for Brain Metastasis Segmentation

Soumya Snigdha Kundu¹ Theodore Barfoot¹ Marina Ivory¹ Kabir Hamzah Muhammad¹
Lorena Garcia-Foncillas Macias¹ Jonathan Shapey^{1,2} Tom Vercauteren¹

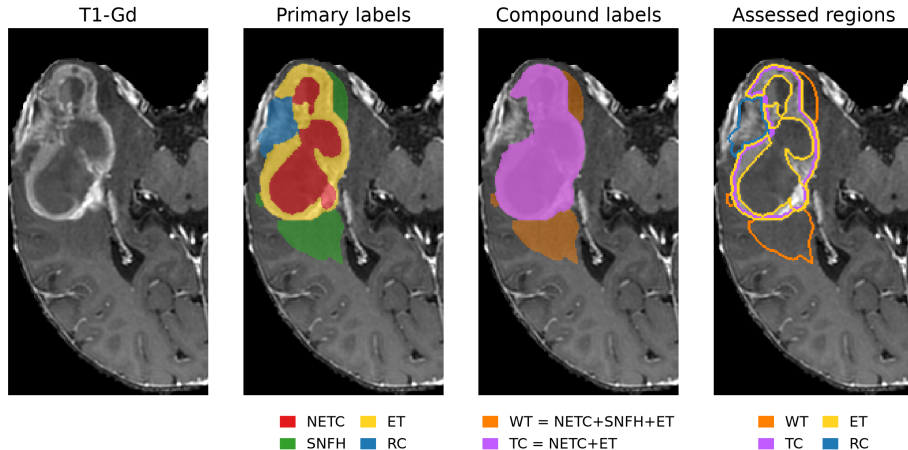
¹School of Biomedical Engineering & Imaging Sciences, King's College London, UK

²Department of Neurosurgery, King's College Hospital NHS Foundation Trust, UK

MICCAI 2026 Challenge BraTS-METS

Presenting Author: Tom Vercauteren

Labels



Training case sub-01321-002 · scored as WT, TC, ET, RC · RC present in only 13% of cases

Metrics

per case and region (■WT ■TC ■ET ■RC) · lesion = connected component · matched 1:1 on any overlap

Lesion-wise DSC

same scheme for NSD

- Large GT lesions only ($\geq 20 \text{ mm}^3$)
- Matched $\rightarrow$ its DSC; missed $\rightarrow 0$
- **Every FP component** adds a 0

$$\text{DSC}_{\text{lw}} = \frac{\sum_{\text{matched}} \text{DSC}_i}{N_{\text{large}} + \text{FP}}$$

Small-lesion F_1

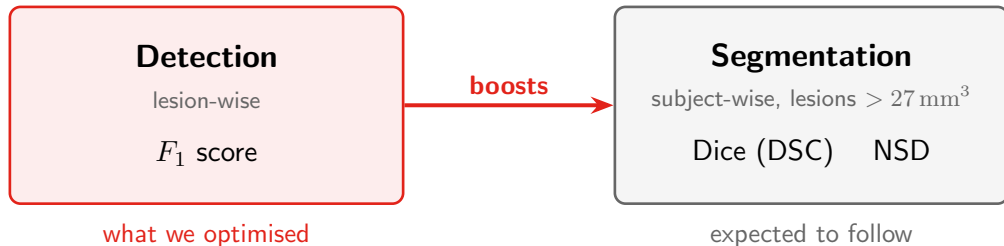
detection

- Small GT lesions only ($< 20 \text{ mm}^3$)
- TP if matched with $\text{DSC} \geq 0.1$, else FN
- FP = **all** unmatched predictions, any size

$$F_1 = \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}}$$

A stray fragment is penalised **twice**: a 0 in the DSC average and an FP in F_1

Hillclimb Goal: Instance Awareness



We optimised for **instance awareness** [1] through the detection task, aiming to **automatically boost segmentation** performance.

Ensemble Candidates

All three: nnU-Net ResEnc-L 3d_fullres, mirroring TTA

Trained on all data: no local validation set created

TopK

4 regions

■ WT ■ TC ■ ET ■ RC

$$\mathcal{L} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{BCE}}^{\text{top}K}$$

$K = 10\% \cdot 1000$ epochs

high-precision base

Completion [1]

6 regions

+ ■ NETC ■ SNFH shells

$$\mathcal{L} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{BCE}}^{\text{top}K}$$

$K = 10\% \cdot 1000$ epochs

decorrelated view

Blob loss [2]

6 regions

Fine-tuned from the final
checkpoint of the Completion run

$$\mathcal{L} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{BCE}}^{\text{top}K} + 0.2 \mathcal{L}_{\text{blob}}$$

100 epochs

high-recall detector

[1] R. Saluja *et al.*, "BackSplit: the importance of sub-dividing the background in biomedical lesion segmentation," CVPR, 2026.

[2] F. Kofler *et al.*, "blob loss: instance imbalance aware loss functions for semantic segmentation," IPMI, 2023.

Training

Setup

- Multi-label: one sigmoid per region, so an  ET voxel is also  TC and  WT
- 6-region:  NETC and  SNFH added as extra channels, dropped at inference
- Decoded in order  WT $\rightarrow$  TC $\rightarrow$  ET $\rightarrow$  RC; inner regions overwrite outer

Design decisions

- Checkpointing based on pseudo-Dice
- Blob loss weight of 0.2 based on [1]
- $K = 10\%$ is the nnU-Net TopK default
- 100 fine-tuning epochs based on a qualitative check of the loss curve

Resection Cavity: Handled Differently

training set, 1295 cases · components $> 27 \text{ mm}^3$

	WT	RC
cases with region	1222	160
single component	31%	85%
2+ components	69%	15%
5+ components	31%	1%
median (max) comps	3 (153)	1 (5)

Tumour: many lesions → **ensemble + hysteresis**

RC: one structure → **single model, keep largest**

- Taken from the **Blob loss model alone**: no ensembling, no hysteresis
- Threshold at 0.5, then keep the **largest component**

official validation set, 179 cases

RC	DSC	NSD	F_1
before	0.581	0.474	0.000
+ RC from Blob loss	+0.049	+0.047	+0.167

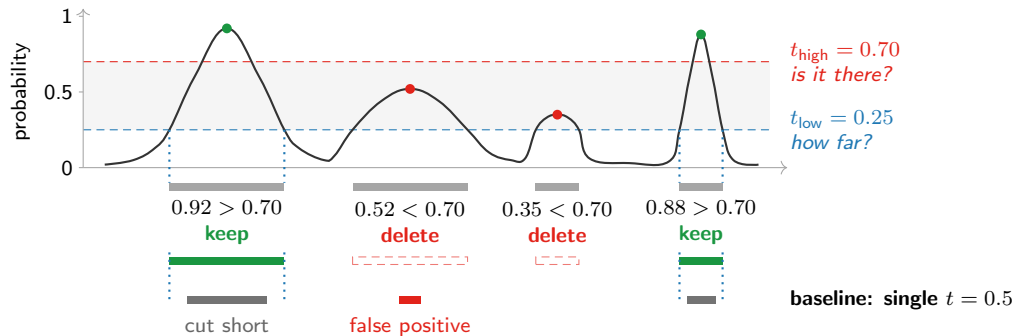
WT / TC / ET unchanged (≤ 0.001)

Hysteresis Decoding

1. Threshold at $t_{\text{low}} = 0.25$ and take the **connected components**
2. For each component: is **any voxel above** $t_{\text{high}} = 0.70$?
 - **No** → **discard** the whole component
 - **Yes** → **keep** the whole component (all its voxels above 0.25)

	Baseline: single $t = 0.5$	Hysteresis
Voxels kept	every voxel > 0.5	voxels > 0.25 , if peak > 0.70
Lesion margins (0.25–0.5)	dropped (cut short)	kept
Weak blob (peak 0.52)	kept (false positive)	discarded

Hysteresis Decoding: Intuition



Primary Ablation

official validation set, 179 cases · each row: change vs. the row above

configuration	DSC				F_1			
	WT	TC	ET	RC	WT	TC	ET	RC
Single model	0.560	0.622	0.593	–	0.379	0.469	0.463	–
+ 2-model ensemble	+0.106	+0.086	+0.091	0.556 [†]	−0.011	−0.030	−0.027	0.000 [†]
+ 2-model ensemble (Completion)	+0.012	+0.010	+0.012	+0.025	+0.002	+0.038	+0.048	0.000
+ RC from Blob loss	+0.001	+0.001	0.000	+0.049	0.000	0.000	0.000	+0.167
+ Hysteresis (0.40 / 0.60)	+0.032	+0.034	+0.033	0.000	−0.005	−0.009	−0.009	0.000
<i>alt.</i> 0.30 / 0.70 [‡]	+0.041	+0.042	+0.041	0.000	+0.006	+0.002	−0.007	0.000
<i>alt.</i> 0.25 / 0.70 [‡]	+0.046	+0.047	+0.050	0.000	+0.006	+0.001	−0.008	0.000
+ 3-model ensemble	+0.009	+0.008	+0.008	0.000	+0.011	+0.011	+0.002	0.000
+ Hysteresis (0.25 / 0.70)	+0.005	+0.005	+0.009	0.000	0.000	−0.001	−0.001	0.000
final	0.725	0.766	0.746	0.630	0.376	0.478	0.476	0.167

[†] first RC score available (single-model RC not recorded) · [‡] alternative thresholds, change vs. the row before hysteresis

Takeaways

- Most prior work acquires instance awareness during training. Our objective was to be instance aware during inference as well.
- Knowing which labels need instance awareness is also key: applied to a label that is usually one structure (RC), instance-aware techniques hurt rather than help.
- There is a clear need to adopt instance-aware evaluation schemes.

Thank you
